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1) $x(t) = \cos(2\pi f_0 t)$, $\quad y(t) = \sum_{k=-\infty}^{\infty} \Pi\left(\frac{t-kT_0}{\frac{T_0}{2}}\right)$

a) Find exponential Fourier series expansion of $x(t)$ and $y(t)$.

b) If $z(t) = x(t)y(t)$ and exponential Fourier series expansion coefficients of $x(t), y(t), z(t)$ are x_n, y_n, z_n respectively, find z_n by using $z_n = \sum_{l=-\infty}^{\infty} x_l\, y_{n-l}$.

c) Sketch $z(t)$ and find its exponential Fourier series expansion directly and compare it to the result in part b).

a) euler: $e^{j\theta} = \cos\theta + j\sin\theta$

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$x(t) = \cos(2\pi f_0 t) = \frac{e^{j2\pi f_0 t} + e^{-j2\pi f_0 t}}{2} = \frac{e^{j2\pi f_0 t}}{2} + \frac{e^{-2\pi f_0 t}}{2}$$

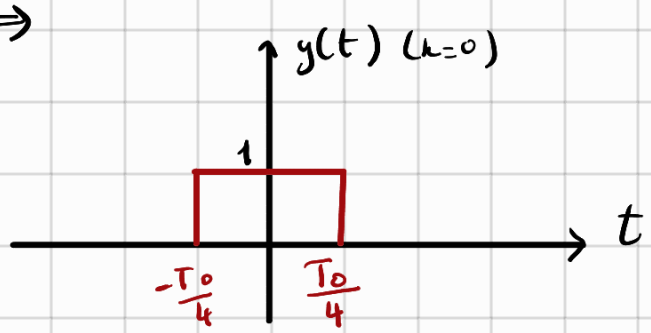
$$x_1 = x_{-1} = \frac{1}{2}$$

Üstel Fs açılımı:

$$x(t) = \sum_{-\infty}^{\infty} x_n e^{jn\omega_0 t} \quad \text{and} \quad x_n = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-jn\omega_0 t}\, dt$$

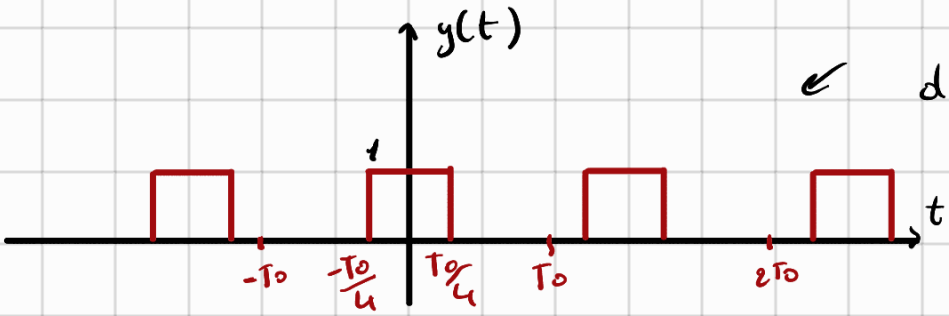
$$y(t) = \sum_{k=-\infty}^{\infty} \Pi\left(\frac{t-kT_0}{\frac{T_0}{2}}\right)$$

$k=0 \Rightarrow$



$$\Pi_2(t/2) = \begin{cases} 1, & -\frac{2}{2} \le t \le \frac{2}{2} \\ 0, & \text{diğer} \end{cases}$$

$\Rightarrow$



darbe katarı

$w_0 = 2\pi f_0 = \frac{2\pi}{T_0}$

$$y_n = \frac{1}{T_0} \int_{T_0} y(t)\, e^{-jnw_0 t}\, dt$$

$$y_n = \frac{1}{T_0} \int_{-T_0/4}^{T_0/4} e^{-jn\frac{2\pi}{T_0} t}\, dt$$

$$y_n = \frac{1}{T_0} \left. \frac{e^{-jn\frac{2\pi}{T_0} t}}{-jn\frac{2\pi}{T_0}} \right|_{-T_0/4}^{T_0/4}$$

$$y_n = \frac{e^{-jn\frac{2\pi}{T_0}\frac{T_0}{4}} - e^{jn\frac{2\pi}{T_0}\frac{T_0}{4}}}{-jn2\pi}$$

$$y_n = \frac{e^{jn\frac{\pi}{2}} - e^{-jn\frac{\pi}{2}}}{jn2\pi} = \frac{\sin(n\frac{\pi}{2})}{n\pi}$$

$\blacktriangleleft \operatorname{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$

$$\Rightarrow y_n = \frac{1}{2} \frac{\sin(n\frac{\pi}{2})}{n\frac{\pi}{2}}$$

$$= \frac{1}{2} \operatorname{sinc}(n/2)$$

$$y(t) = \sum_{n=-\infty}^{\infty} y_n e^{jnw_0 t}$$

$$w_0 = \frac{2\pi}{T_0}$$

$$\boxed{y(t) = \sum_{n=-\infty}^{\infty} \frac{1}{2} \operatorname{sinc}(n/2) e^{jn\frac{2\pi}{T_0}t}}$$

$$= \sum_{n=-\infty}^{\infty} \frac{1}{2} \operatorname{sinc}(n/2) e^{jn2\pi f_0 t}$$

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b)

$$x(t) = \frac{e^{j2\pi f_0 t}}{2} + \frac{e^{-2\pi f_0 t}}{2}$$

$x_{-1} = x_1 = \frac{1}{2}$, others $x_n = 0$

$$y(t) = \sum_{n=-\infty}^{\infty} \frac{1}{2} \operatorname{sinc}(n/2) e^{jn2\pi f_0 t}$$

$$y_{n+1} = \frac{1}{2} \operatorname{sinc}\left(\frac{n+1}{2}\right)$$

$$y_{n-1} = \frac{1}{2} \operatorname{sinc}\left(\frac{n-1}{2}\right)$$

$$z_n = \sum_{\ell=-\infty}^{\infty} x_\ell\, y_{n-\ell}$$

$$z_n = x_{-1}\, y_{n+1} + x_1\, y_{n-1}$$

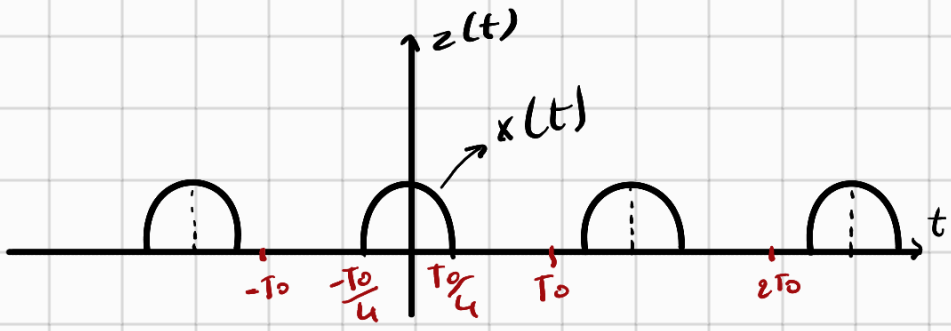
$$= \frac{1}{2}\left[\frac{1}{2} \operatorname{sinc}\left(\frac{n+1}{2}\right) + \frac{1}{2} \operatorname{sinc}\left(\frac{n-1}{2}\right)\right]$$

$$= \frac{1}{4}\left[\operatorname{sinc}\left(\frac{n+1}{2}\right) + \operatorname{sinc}\left(\frac{n-1}{2}\right)\right]$$

$$z(t) = \sum_{n=-\infty}^{\infty} z_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{\infty} z_n e^{j2\pi f_0 t}$$

$$= \sum_{n=-\infty}^{\infty} \frac{1}{4}\left[\operatorname{sinc}\left(\frac{n+1}{2}\right) + \operatorname{sinc}\left(\frac{n-1}{2}\right)\right] e^{j2\pi f_0 t}$$

c)



Periyod = T_0 havevalue between $-\frac{T_0}{4} \leq T_0 \leq \frac{T_0}{4}$

$z(t) = x(t) = \cos(2\pi f_0 t)$ when

$$z_n = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-jn\omega_0 t}\, dt$$

$$= \frac{1}{T_0}\int_{-T_0/4}^{T_0/4} \frac{e^{j2\pi f_0 t} + e^{-j2\pi f_0 t}}{2} \cdot e^{-jn2\pi f_0 t}\, dt$$

$$= \frac{1}{2T_0}\int_{-T_0/4}^{T_0/4} e^{j\frac{2\pi}{T_0}t(1-n)} + e^{-j\frac{2\pi}{T_0}t(1+n)}\, dt$$

$$= \frac{1}{2T_0}\left[\frac{e^{j\frac{2\pi}{T_0}\frac{T_0}{4}2(1-n)}}{j\frac{2\pi}{T_0}(1-n)} - \frac{e^{-j\frac{2\pi}{T_0}\frac{T_0}{4}2(1-n)}}{j\frac{2\pi}{T_0}(1-n)} + \right.$$

$$\left. \frac{e^{-j\frac{2\pi}{T_0}\frac{T_0}{4}2(1+n)}}{-j\frac{2\pi}{T_0}(1+n)} - \frac{e^{+j\frac{2\pi}{T_0}\frac{T_0}{4}2(1+n)}}{-j\frac{2\pi}{T_0}(1+n)} \right]$$

$$= \frac{1}{2}\left[\frac{e^{j\frac{\pi}{2}(1-n)} - e^{-j\frac{\pi}{2}(1-n)}}{j2\pi(1-n)} + \frac{e^{j\frac{\pi}{2}(1+n)} - e^{-j\frac{\pi}{2}(1+n)}}{j2\pi(1+n)}\right]$$

$$= \frac{1}{2}\left[\frac{\sin\left(\frac{\pi}{2}(1-n)\right)}{\pi(1-n)} + \frac{\sin\left(\frac{\pi}{2}(1+n)\right)}{\pi(1+n)}\right]$$

$$= \frac{1}{4}\left[\frac{\sin\left(\frac{\pi}{2}(1-n)\right)}{\frac{\pi}{2}(1-n)} + \frac{\sin\left(\frac{\pi}{2}(1+n)\right)}{\frac{\pi}{2}(1+n)}\right]$$

$$= \frac{1}{4}\left[\underbrace{\operatorname{sinc}\left(\frac{1-n}{2}\right)}_{\text{çift}} + \operatorname{sinc}\left(\frac{1+n}{2}\right)\right]$$

$$= \frac{1}{4}\left[\operatorname{sinc}\left(\frac{n-1}{2}\right) + \operatorname{sinc}\left(\frac{n+1}{2}\right)\right] = z_n$$

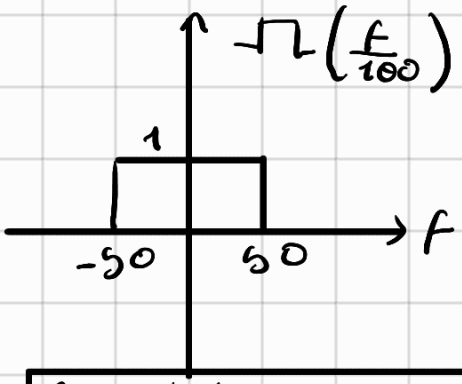
$$\text{and } z(t) = \sum_{n=-\infty}^{\infty} \frac{1}{4}\left[\operatorname{sinc}\left(\frac{n+1}{2}\right) + \operatorname{sinc}\left(\frac{n-1}{2}\right)\right] e^{j2\pi f_0 t}$$

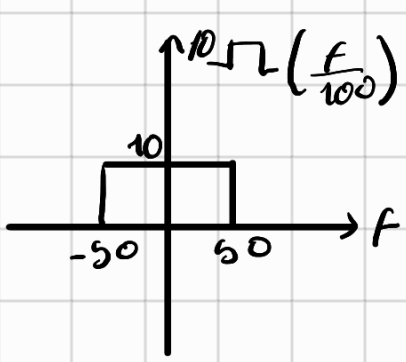
The same at b//

2) $\underbrace{Y(f)}$ $\underbrace{Z(f)}$

a) If $X(f) = 10\Pi\left(\frac{f}{100}\right) * \Pi\left(\frac{f}{100}\right)$, find $X(f)$ by using convolution integral.

b) Find $x(t)$ by using inverse Fourier transform. Is $x(t)$ energy or power signal? Find its power or energy content.



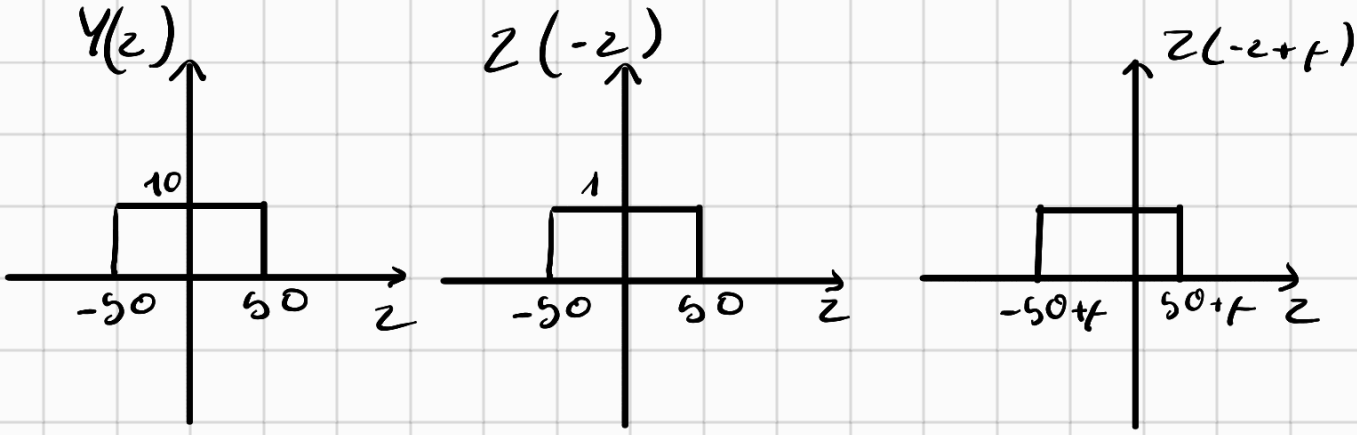


$$\Pi_2(t/2) = \begin{cases} 1, & -\frac{2}{2} \le t \le \frac{2}{2} \\ 0, & \text{diğer} \end{cases}$$

Convulation

$$x(t) * y(t) = \int_{-\infty}^{\infty} x(z)\, y(t-z)\, dz$$

$$X(f) = \int_{-\infty}^{\infty} 10\,\Pi\left(\frac{z}{100}\right)\Pi\left(\frac{f-z}{100}\right)dz$$



if $f < 100$ $X(f) = 0$

if $-100 < f < 0$ $X(f) = \int_{-50}^{50+f} 10 \cdot 1\,dz$

$= 10z\Big|_{-50}^{50+f} = 500 + 10f + 500$

$= 1000 + 10f$

if $0 < f < 100$ $X(f) = \int_{-50+f}^{50} 10 \cdot 1\,dz$

$= 10z\Big|_{-50+f}^{50} = 500 - (-500 + 10f)$

$= 1000 - 10f$

if $f > -100$ $X(f) = 0$

so $X(f) = \begin{cases} 1000\,(1 + f/100), & \text{if } -100 < f < 0 \\ 1000\left(1 - \frac{f}{100}\right), & \text{if } 0 < f < 100 \\ 0, & \text{else} \end{cases}$

$$\wedge\left(\frac{t}{2}\right) = \begin{cases} 1 - \frac{|t|}{2}, & \frac{|t|}{2} \le 1 : |t| \le 2 \\ 0, & \text{else} \end{cases}$$

$$X(f) = 1000\,\underbrace{\wedge\left(\frac{f}{100}\right)}_{Z(f)}$$

b) $y(t) = \Lambda(t/2) \xleftrightarrow{F} Y(f) = 2\,\mathrm{sinc}^2(f2)$

$Y(t) = 2\,\mathrm{sinc}^2(t2) \xleftrightarrow{F} y(-f) = \Lambda\left(\frac{+f}{2}\right)$ (simetrik)

$Z(f) = \Lambda\left(\frac{f}{100}\right) \xleftrightarrow{F} z(t) = 100\,\mathrm{sinc}^2(100t)$

$X(t) = 100\,000\,\mathrm{sinc}^2(100t)$

Parseval theorem

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f^2)|^2 df$$

$$= \int_{-\infty}^{\infty} \left|1000\,\Lambda\left(\frac{f}{100}\right)\right|^2 df$$

$$= \int_{-100}^{100} \left|1000\left(1-\frac{|f|}{100}\right)\right|^2 df$$

$$= \int_{-100}^{0} \left|1000\left(1+\frac{f}{100}\right)\right|^2 df + \int_{0}^{100} \left|1000\left(1-\frac{f}{100}\right)\right|^2 df$$

$$= (1000)^2 \left[\int_{-100}^{0} df + \int_{-100}^{0} \frac{f^2}{100^2} df + \int_{-100}^{0} \frac{2f}{100}\right]$$

$$+ (1000)^2 \left[\int_{0}^{100} df + \int_{0}^{100} \frac{f^2}{100^2} df - \int_{0}^{100} \frac{2f}{100}\right]$$

$$= (1000)^2 \left[100 + \frac{f^3}{3\cdot 100^2}\Big|_{-100}^{0} + \frac{2f^2}{2\cdot 100}\Big|_{-100}^{0} + 100 + \frac{f^3}{3\cdot 100^2}\Big|_{0}^{100} - \frac{2f^2}{2\cdot 100}\Big|_{0}^{100}\right]$$

$$= (1000)^2 \left[100 - \frac{(-100)^3}{3\cdot 100^2} - \frac{(100)^2}{100} + 100 + \frac{(100)^3}{3\cdot 100^2} - \frac{(100)^2}{100}\right]$$

$$= (1000)^2 \left[\frac{100}{3} + \frac{100}{3}\right]$$

$$= (1000)^2 \left[\frac{200}{3}\right] = 66{,}6\cdot 10^6\,\mathrm{J} = 66{,}6\,\mathrm{MJ}$$

3) If $x(t) = \begin{cases} cos(2\pi f_0 t) & , \; -\frac{T_0}{2} \le t \le \frac{T_0}{2} \\ 0 & , \; Else \end{cases}$, find $X(f)$ and sketch it.

$$x(t) = \cos(2\pi f_0 t) \underbrace{\Pi(t/T_0)}_{z(t)}$$

$$y(t) = A\,\Pi(t/2) \overset{F}{\longleftrightarrow} Y(f) = A\,2\,\mathrm{sinc}(f2)$$

$$z(t) = \Pi(t/T_0) \overset{F}{\longleftrightarrow} Z(f) = T_0\,\mathrm{sinc}(T_0 f)$$

Modülasyon $y(t) = x(t)\cos(2\pi f_0 t) \overset{F}{\longleftrightarrow} \frac{1}{2}[X(f-f_0) + X(f+f_0)]$

$$X(f) = \frac{T_0}{2}\left[\mathrm{sinc}\left(T_0(f-f_0)\right) + \mathrm{sinc}\left(T_0(f+f_0)\right)\right]$$

$$X(f) = \frac{T_0}{2}\left[\mathrm{sinc}\left(T_0 f - 1\right) + \mathrm{sinc}\left(T_0 f + 1\right)\right]$$

$\mathrm{sinc}(t) = \frac{\sin \pi t}{\pi t}$

4) If $x(t) = \frac{1}{t}$, find $X(f)$ and sketch it. (Hint: Use duality property.)

Duality property

$x(t) \xleftrightarrow{F} X(f)$

$X(t) \xleftrightarrow{F} x(-f)$

$y(t) = \operatorname{sgn}(t) \xleftrightarrow{F} Y(f) = \frac{1}{j\pi f}$

$Y(t) = \frac{1}{j\pi t} \xleftrightarrow{F} y(-f) = \operatorname{sgn}(-f)$

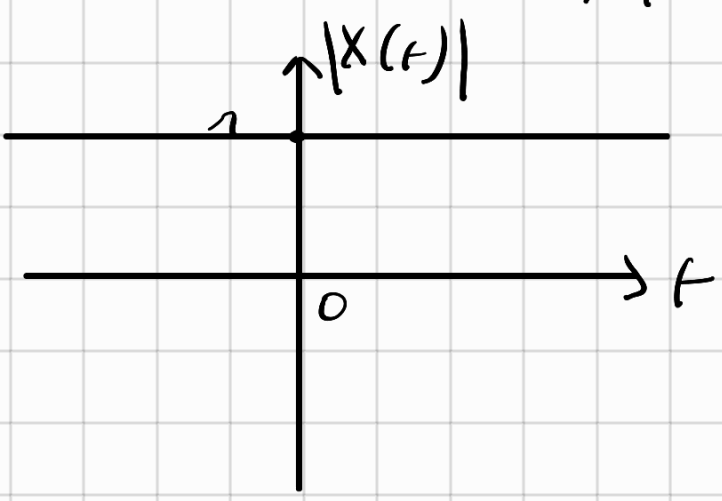
$x(t) = Y(t) \cdot j\pi \xleftrightarrow{F} X(f) = y(-f) \cdot j\pi = j\pi \operatorname{sgn}(-f)$

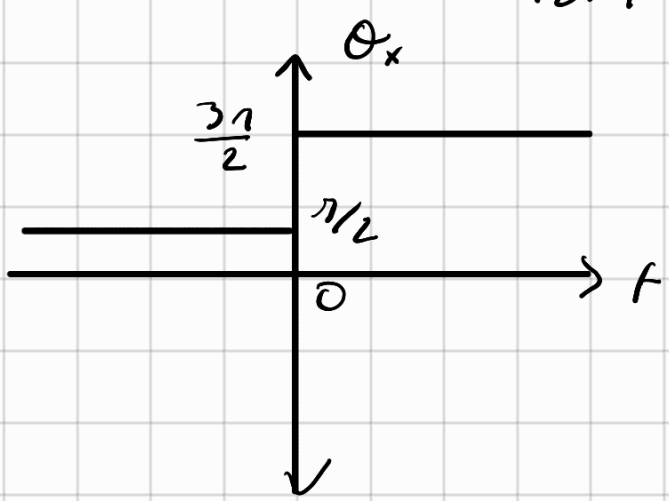
$= \pi e^{j\pi/2} \operatorname{sgn}(-f)$

$$\operatorname{sgn}(-f) = \begin{cases} -1, & f > 0 \\ +1, & f < 0 \end{cases} \quad (-1 = e^{j\pi})$$

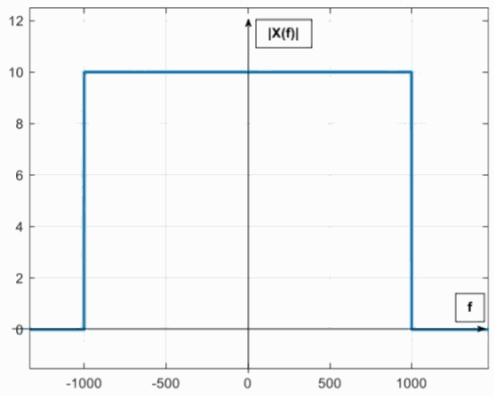
$$X(f) = \begin{cases} \pi e^{j\pi} \cdot e^{j\pi/2}, & f > 0 \\ \pi e^{j\pi/2}, & f < 0 \end{cases}$$

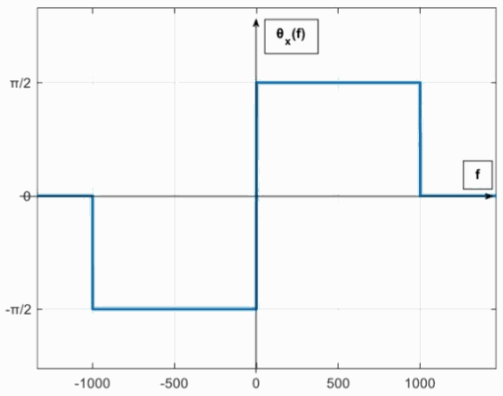
$$= \begin{cases} \pi e^{3\pi/2}, & f > 0 \\ \pi e^{j\pi/2}, & f < 0 \end{cases} \Rightarrow |X(f)| = \pi, \quad \theta_x = \begin{cases} 3\pi/2, & f > 0 \\ \pi/2, & f < 0 \end{cases}$$





5)





By using inverse Fourier transform, find $x(t)$.

$$X(f) = \begin{cases} 10\, e^{-j\pi/2}, & -1000 < f < 0 \\ 10\, e^{j\pi/2}, & 0 < f < 1000 \\ 0, & \text{else} \end{cases}$$

$$x(t) = \int_{-\infty}^{\infty} X(f)\, e^{j2\pi ft}\, df$$

$$= \int_{-1000}^{0} 10\, e^{-j\pi/2}\, e^{j2\pi ft}\, df + \int_{0}^{1000} 10\, e^{j\pi/2}\, e^{j2\pi ft}\, df$$

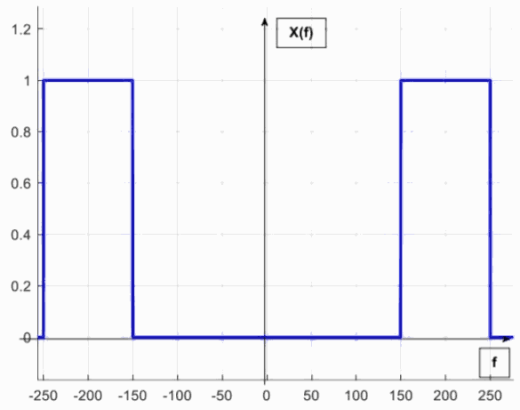
$$= 10 e^{-j\pi/2} \int_{-1000}^{0} e^{j2\pi ft}\, df + 10 e^{j\pi/2} \int_{0}^{1000} e^{j2\pi ft}\, df$$

$$= 10 e^{-j\pi/2} \left. \frac{e^{j2\pi ft}}{j2\pi t} \right|_{-1000}^{0} + 10 e^{j\pi/2} \left. \frac{e^{j2\pi ft}}{j2\pi t} \right|_{0}^{1000}$$

$$= -10j \left[\frac{1 - e^{-j2000\pi t}}{j2\pi t} \right] + 10j \left[\frac{e^{j2000\pi t} - 1}{j2\pi t} \right]$$

$$= \frac{-10 + 10\cos(2000\pi t)}{\pi t}$$

6)



By using inverse Fourier transform, find $x(t)$.

$$x(t) = \int_{-\infty}^{\infty} X(f)\, e^{j2\pi ft}\, df$$

$$= \int_{-250}^{-150} e^{j2\pi ft}\, df + \int_{150}^{250} e^{j2\pi ft}\, df$$

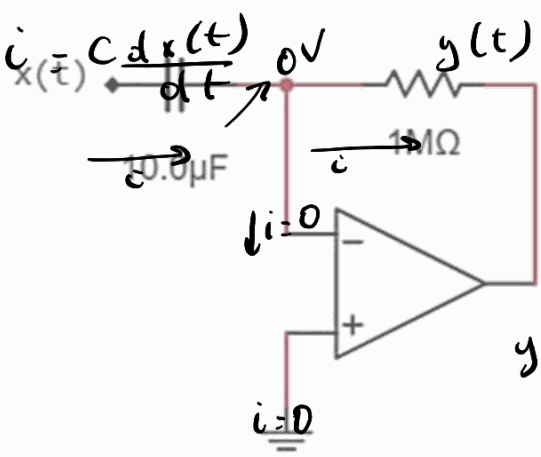
$$= \frac{e^{j2\pi ft}}{j2\pi t}\Big|_{-250}^{-150} + \frac{e^{j2\pi ft}}{j2\pi t}\Big|_{150}^{250}$$

$$= \frac{e^{-j300\pi t} - e^{-j500\pi t} + e^{j500\pi t} - e^{j300\pi t}}{j2\pi t}$$

$$= -\frac{\sin(300\pi t)}{\pi t} + \frac{\sin(500\pi t)}{\pi t}$$

$$= -300\,\mathrm{sinc}(300t) + 500\,\mathrm{sinc}(500t)$$

7)



$$y(t) = -\frac{C\,dx(t)}{dt} \cdot 1M$$

$$= -\frac{10\mu F\,dx(t)}{dt}\,1M$$

$$y(t) = \frac{-10\,dx(t)}{dt}$$

If the power spectral density (PSD) of the input voltage $x(t)$ is $S_x(f) = 10\Lambda\left(\frac{f}{100}\right)$, find the power of the output voltage $y(t)$.

$$S_x(f) = 10\,\Lambda\left(\frac{f}{100}\right) = \lim_{T\to\infty} \frac{1}{T}|X(f)|^2$$

$$Y(f) = -10\,j2\pi f\,X(f)$$

$$S_y(f) = \lim_{T\to\infty} \frac{1}{T}\left|-10\,j2\pi f\,X(f)\right|^2$$

$$S_y = \lim_{T\to\infty} \frac{1}{T}\left|-10\,j2\pi f\right|^2 \left|X(f)\right|^2$$

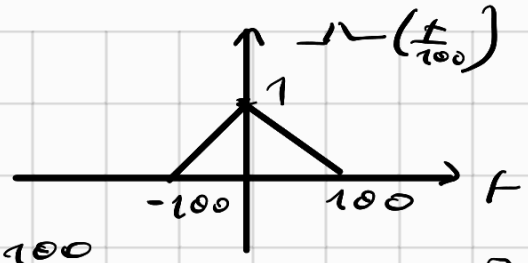
$$= \left|-10\,j2\pi f\right|^2 \lim_{T\to\infty} \frac{1}{T}|X(f)|^2$$

$$S_y(f) = \left|-10\,j2\pi f\right|^2 S_x(f)$$

$$S_y(f) = 100\,(2\pi f)^2\,10\,\Lambda\left(\frac{f}{100}\right)$$

$$= +4000\pi^2 f^2\,\Lambda\left(\frac{f}{100}\right)$$

$$\Rightarrow P_{y(t)} = \int_{-\infty}^{\infty} S_y(f)\, df$$



$$= +4000\pi^2 \left[\int_{-100}^{0} f^2 \left(\frac{+f}{100} + 1 \right) df + \int_{0}^{100} f^2 \left(\frac{-f}{100} + 1 \right) df \right]$$

$$= +4000\pi^2 \left[\int_{-100}^{0} \frac{f^3}{100} df + \int_{-100}^{0} f^2\, df + \int_{0}^{100} \frac{-f^3}{100} + \int_{0}^{100} f^2 df \right]$$

$$= +4000\pi^2 \left[\frac{f^4}{400}\Big|_{-100}^{0} + \frac{f^3}{3}\Big|_{-100}^{0} - \frac{f^4}{400}\Big|_{0}^{100} + \frac{f^3}{3}\Big|_{0}^{100} \right]$$

$$= 4000\pi^2 \left[-\frac{(-100)^4}{400} + \frac{(+100)^3}{3} - \frac{(100)^4}{400} + \frac{(100)^3}{3} \right]$$

$$= 4000\pi^2 \left[166{,}66 \cdot 10^3 \right]$$

$$= 666{,}6 \cdot 10^6 \pi^2 = 6{,}5797 \cdot 10^9 = 6{,}5797\,\text{TW}$$